

Multiple Choice

1. **Answer:** A

Options: A) `**`; B) `//`; C) `is`; D) not in

Note: The operator `**` in Python 3 is specifically designed to perform exponentiation, allowing users to raise a number to the power of another number.

2. **Answer:** A

Options: A) $O(1)$; B) $O(n)$; C) $O(n^2)$; D) $O(\log n)$

Note: $O(1)$ is the smallest asymptotic notation because it represents a constant time complexity that does not grow with the size of the input, indicating that the algorithm's performance remains unchanged regardless of the input size.

3. **Answer:** D

Options: A) A function that averages numeric values in a column or row; B) A function that counts the values in a column or row; C) A function that rounds a numeric value; D) A function that sorts values in a column or row

Note: Sorting values in a column or row allows users to easily identify outliers or discrepancies, making it easier to detect improbably high or low values that may indicate data entry errors.

4. **Answer:** D

Options: A) Error; B) a; C) b; D) c

Note: In Python 3, negative indexing allows access to elements from the end of a string, so `"abc"[-1]` refers to the last character, which is `'c'`.

5. **Answer:** C

Options: A) Error; B) abc; C) cba; D) c

Note: The output of `"abc"[::-1]` is `"cba"` because the slicing syntax `[::-1]` reverses the string by stepping through it backward.

True / False

6. **Answer:** False

Options: A) True; B) False

Note: The sum of the list `l = [1, 2, 3, 4]` in Python 3 is equal to 10, not 8, because when you add the numbers together ($1 + 2 + 3 + 4$), the total is 10.

7. **Answer:** False

Options: A) True; B) False

Note: The expression `3 % 3` evaluates to 0, so $1 + 0$ equals 1, not 3, making the statement false.

8. **Answer:** False

Options: A) True; B) False

Note: The expression `b[::2]` returns `[11, 15, 19]` from the list `b = [11, 13, 15, 17, 19, 21]`, as it selects every second element starting from the first index.

9. **Answer:** True

Options: A) True; B) False

Note: Even if a large Java program passes all tests without errors, it may still contain bugs that were not encountered during testing due to limitations in test coverage or untested edge cases.

10. **Answer:** True

Options: A) True; B) False

Note: The `isupper()` function in Python 3 returns True if all the characters in a string are uppercase letters and there is at least one character, confirming that it checks for the uppercase status of all characters in the string.

Long Form Response

11. **Answer:** `def mul_consecutive_nums(nums): | return result`

Note: The answer correctly identifies the need for a function that processes a list of numbers to compute the product of consecutive elements, which is essential for fulfilling the task of multiplying those numbers together.

12. **Answer:** `def max_difference(test_list): | return (res)`

Note: The answer correctly identifies the need to create a function that calculates and returns the maximum difference between pairs in a list of tuples, though it lacks the implementation details for the actual calculation.

End of Answer Key